

Chandula Adhikari

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Research Interests

My research interests lie at the intersection of **computational biology** and **deep learning**, with a focus on **single-cell genomics**, **gene regulation**, and **generative models** for biomedical data. I am particularly interested in models that are built around the underlying biology and whose assumptions remain interpretable.

Education

University of Peradeniya

BSc.Eng(Hons.) in Computer Engineering, **First Class**

Mar. 2021 - Aug. 2025

Overall GPA: **3.85/4.0**

Final-Year GPA: **4.00/4.00**

Dharmaraja College, Kandy

G.C.E. Advanced Level Examination - 467 from 19500+ participants nationwide

Jan. 2006 - Dec. 2019

Z-Score: **2.2069**

Publications

TARVI: Transcription-Factor Aided RNA Velocity Inference with Supervised Latent Time | 🌐 | 🌐

R.A.J.C. Adhikari, S. Dassanayake, D. Herath

- Replaced the static per-gene transcription rate used by existing RNA velocity models with **cell-specific, TF-regulated rates**, injecting a masked learnable TF→gene matrix derived from **ENCODE ChIP-seq and ChEA** regulatory priors into a variational autoencoder, a knowledge-guided constraint that makes the model's regulatory assumptions explicit and inspectable.
- Introduced a supervised latent-time head trained with ODE-residual, self-distillation and graph-Laplacian smoothness losses, plus post-hoc velocity–pseudotime blending that grounds local velocity vectors in global trajectory topology.
- Improved pseudotime calibration by **126%** over VeloVI (Spearman 0.747 vs. 0.331) across 5 single-cell datasets (pancreas, bone marrow, forebrain, chromaffin, scEU organoid), while matching the strongest of 4 baselines on directional metrics.
- Best Paper Award**, Biomedical Engineering & Instrumentation track | MERCon 2026, pp. 922–927 | IEEE Xplore

Dropout Is the Absorbing State: Discrete Diffusion for Single-Cell Gene Expression Imputation

Under review, AAAI 2027

R.A.J.C. Adhikari et al.

- Reformulated scRNA-seq dropout as the **absorbing state of a discrete diffusion process**: expression is tokenized into per-gene quantile bins and only candidate dropout entries are corrupted toward a reserved [MASK] state, so measured entries stay immutable, non-negativity holds by construction, and each gene receives a full categorical distribution rather than a conditional mean.
- Designed a role-conditioned Transformer denoiser and a **confidence-first parallel decoding** scheme that exploits the monotonicity of the absorbing chain to reconstruct the matrix in a few passes instead of a full ancestral trajectory.
- Against **9 baselines** (MAGIC, scImpute, SAVER, ALRA, DCA, scVI, scIGANs, scIDPMs) on 4 simulated datasets at 50–72% dropout, achieved the best rank correlation (ρ 0.90–0.94) and lowest log-scale RMSE (0.55–0.61) on every dataset, and won **20 of 21** cell-type clustering views across 3 real datasets.

RCT-PEARL: Annotated Periapical Radiograph Dataset for AI-Based Quality Assessment of Root Canal Treatment | *Under review, Nature Scientific Data*

S. Rasnayaka, R.M. Jayasinghe, A. Hewage, M. Fonseka, C. Adhikari, et al.

- A public dataset of **1,311 anonymized intraoral periapical radiographs** of root canal treated teeth, with region annotations for treated teeth and filling instances, root apex keypoints, and per-filling quality labels for filling length, lateral seal, voids, and irregularities.

Uncovering Hidden Temporal Patterns in T20I Cricket Through Ball-by-Ball Data Analysis

R.A.J.C. Adhikari et al.

- Analyzed 165,000+ deliveries from 1,200+ T20I matches (2006–2025) to identify temporal patterns in batting and bowling performance across powerplay, middle, and death overs.
- ICATC 2025 | IEEE Xplore

Optimized Multi-Processor System-on-Chip (MPSoC) Design for Low-Resource JPEG Encoding

K.H. Gunawardana, R.A.J.C. Adhikari, I. Nawinne

- Proposed a pipelined MPSoC architecture for efficient JPEG encoding using Altera Nios II/e cores on a Cyclone IV FPGA, integrating custom instructions, FIFO-based inter-core communication, and superscalar enhancements to maximize throughput.
- ICAC 2024 | IEEE Xplore

Experience

Research Assistant | *Multidisciplinary AI Research Centre (MARC), University of Peradeniya* June 2026 – Present
Research post at the University of Peradeniya’s interdisciplinary AI centre, which coordinates machine learning across engineering, medicine and the life sciences.
Lead two research groups at the centre: the **Computational Biology & Bioinformatics** pillar, and **Generative AI for Dermatology**.
Work carried out in this post produced the absorbing-state discrete diffusion imputation model currently **under review at AAAI 2027**.

Temporary Instructor | *Department of Computer Engineering* Sep. 2025 – June 2026
Conducted laboratory sessions and prepared tutorials and lab specifications for undergraduate courses; supervised and mentored student project work. Built on three prior years as an Undergraduate Teaching Assistant (Nov. 2023 – July 2025) across Data Structures & Algorithms, Computer Architecture, Digital Design and Embedded Systems.

Intern Data Scientist | *OCTAVE, John Keells Holdings* July 2024 – Dec. 2024
Built and validated production-grade data pipelines over large enterprise datasets, and developed customer segmentation models that directly supported live digital marketing campaigns, gaining experience in end-to-end data science workflow, reproducibility and experiment tracking that transfer to large-scale genomic data analysis.
Technologies: Databricks, PySpark, MLflow, Azure DevOps

Selected Projects

AI-Assisted Tool for Assessing Quality of Final Root Canal Treatment Using Dental IOPA Radiographs | *Group* |  Jan. 2025 – Present

- Developing an explainable AI framework for automated root canal treatment (RCT) quality assessment using 1,000 anonymized intraoral periapical radiographs. The four-stage pipeline (dataset curation, expert-guided annotation, segmentation, and multi-parameter evaluation) enables pixel- and geometry-based analysis of clinical parameters such as filling length, lateral seal, voids, and irregularities.
- Contribution:** Dataset preparation, annotation design, quality parameter evaluation, and explainable output visualization.
- Technologies:** Python, PyTorch, YOLO, OpenCV, Keras.

Reliability of RNA Velocity in Low-Dimensional Projections | *Individual* Feb. 2025 – Dec. 2025

- Benchmarked custom autoencoder architectures against standard embeddings (PCA, t-SNE, UMAP) across 4 biological datasets, showing that denoising autoencoders better preserve local neighborhood structure and velocity-field coherence, establishing that the embedding, not the velocity estimator alone, limits trajectory inference.
- Findings motivated and fed directly into **TARVI** (MERCon 2026, Best Paper Award), which addressed the resulting calibration gap at the model level.
- Technologies:** Denoising Autoencoders, VAEs | PCA, t-SNE, UMAP | scVelo (dynamical model), Velocityto | Evaluation: k-NN preservation, distance correlation, Jaccard similarity

BeeZee: Smart Beehive Monitoring System | *Group* |  |  Nov. 2023 - July 2024

- Collaborative research with the Faculty of Agriculture, University of Peradeniya to detect early signs of bee colony abscondment. Designed and built a cloud-connected data collection unit with sensors (CO₂, humidity, temperature, weight) and camera, integrated into a real-time dashboard. Successfully implemented pollen-carrying bee detection using YOLOv8 and StrongSORT for object tracking.
- Contribution:** Hardware enclosure, circuit assembly, sensor calibration, the front-end dashboard, and the computer vision pipeline for bee detection and tracking.
- Technologies:** Raspberry Pi, AWS(S3, IoT Core, Lambda), Node.js, MongoDB, YOLOv8, StrongSORT.

Technical Skills

Languages	Python, C/C++, Java, JavaScript/TypeScript, Verilog HDL, ARM Assembly
Machine & Deep Learning	PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face Transformers
Model Classes	Transformers, Diffusion Models (DDPM/DDIM, absorbing-state discrete diffusion), VAEs & Autoencoders, CNNs, ODE-constrained neural models
Single-Cell & Bioinformatics	Scanpy, AnnData, scVelo, Velocityto, Seurat; ENCODE ChIP-seq & ChEA regulatory atlases; trajectory inference, dropout imputation, cell-type clustering
Data & Cloud	NumPy, Pandas, Matplotlib, Seaborn, OpenCV; Databricks, PySpark, SQL (Spark SQL & T-SQL), MLflow, Azure DevOps, AWS, Power BI
Developer Tools	Git/GitHub, Unix Shell, Linux/HPC, VS Code, PyCharm/IntelliJ, L ^A T _E X, Jekyll, Jenkins
Web Frameworks	React.js, Express.js, Flutter, Spring-Boot, FastAPI

Achievements / Competitions

*Algorithmic programming contests, all entered with team **Five4Five**:*

IEEE Xtreme 17.0 24-hour global Global Rank 174 of 7,091	Nov. 2023
MoraXtream 8.0 12-hour, IEEE Student Branch, Univ. of Moratuwa National Rank 4 of 450+	Nov. 2023
Aces Coders v10.0 12-hour, ACES, Univ. of Peradeniya National Rank 8 of 350+	Oct. 2023
Aces PreCoders v10.0 6-hour Rank 2 of 50+	Sep. 2023
NBQSA National ICT Awards IntelliSwitcher: Intelligent Domestic Energy Optimizing System 	Aug. 2023

Selected Certificates

Certificate Course in Molecular Biology & Biotechnology | 100 hours 2026

Agricultural Biotechnology Centre, University of Peradeniya

- Formal wet-lab and bioinformatics training taken to ground my computational work in experimental biology: **gene expression regulation and mechanisms**, next-generation and Sanger sequencing, genome organization, recombinant DNA cloning, DNA/RNA extraction, PCR, and gel electrophoresis; bioinformatics modules on NCBI resources, primer design, **Nanopore transcriptomics analysis**, and molecular phylogenetics. (40h theory, 40h practical, 20h assessed independent work.)

Machine Learning Specialization | Stanford University and DeepLearning.AI(Coursera) Apr. 2024

- Supervised Machine Learning: Regression and Classification
- Advanced Learning Algorithms

Python for Data Science, AI and Development | IBM and Coursera Mar. 2023

References

All three referees are at the University of Peradeniya, Sri Lanka.

Prof. Roshan G. Ragel | roshanr@eng.pdn.ac.lk

Professor, Department of Computer Engineering

Dr. Damayanthi Herath | damayanthiherath@eng.pdn.ac.lk

Senior Lecturer, Department of Computer Engineering; Coordinator, Computational Biology (CompBio) Group

Prof. G. M. Roshan I. Godaliyadda | roshang@eng.pdn.ac.lk

Professor, Department of Electrical & Electronic Engineering

Deputy Director (Research & Innovation), MARC